SMART INDIA HACKATHON 202

TITLE PAGE

- **Problem Statement ID – SIH26040**
- **Problem Statement Title- FloatChat: AI-Powered Conversational Interface for ARGO Ocean Data Discovery and Visualization**
- **Theme- Miscellaneous (Ocean Technology)**
- **PS Category- Software**
- **Team ID- (To be added after registration)**
- **Team Name- Ctrl Alt Elites**

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IDEA TITLE

❖ Proposed Solution

Idea Title: FloatChat — "Ocean Se Baat Karo": Voice + AI chatbot for 4000+ Argo floats, on WhatsApp, in any Indian language

Solution

- Voice (any Indian language) + text (English/Hindi) interface to query ocean data
- Query → LLM converts to SQL + spatial filter → Argo DB → 3D depth view + map + voice explanation
- AnomalyRadar — proactively alerts on critical thresholds (cyclone precursor, marine heatwave)
- Delivery: Web dashboard + WhatsApp/Telegram bot — zero app download, zero training

How It Addresses the Problem

- NetCDF-locked data → converted to SQL + vector DB, opened up for everyone; voice/chat removes all technical barriers
- WhatsApp delivery reaches 500M+ Indian users — no new app needed; proactive alerts, not just answers on demand

Innovation & Uniqueness — vs Argovis, Euro-Argo, ODV, SAMUDRA

Feature	Existing Tools	FloatChat
OceanVoice — Hindi/Tamil/Telugu voice query & response	None offer this	✓
AnomalyRadar — proactive cyclone/heatwave alerts	Reactive-only	✓
WhatsApp/Telegram Bot — query on existing apps	No interactive bots	✓
OceanLens 3D — WebGL depth cross-sections	2D plots only	✓
FloatChat Classroom — adopt a real Argo float	No education mode	✓

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TECHNICAL APPROACH

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Frontend	React + TS + Tailwind, Recharts + Three.js/Deck.gl (3D), Leaflet maps
Voice Engine	Web Speech API (browser-native) + Whisper fallback
Backend	FastAPI (Python) — query engine + LLM orchestration + anomaly detection
Database	PostgreSQL + FAISS/Chroma (vector DB for RAG)
AI/LLM	LLaMA 3 / Mistral via OmniRoute — NL→SQL + RAG + insight generation
Data Pipeline	Python (xarray + netCDF4 + argopy) — NetCDF → PostgreSQL + Parquet
Delivery	Web App + WhatsApp Business API + Telegram Bot API

DATA INGESTION LAYER
ARGO GDAC/INCOIS NetCDF → xarray/argopy → PostgreSQL + Parquet + FAISS (auto-sync every 6 hrs)

USER INTERFACE LAYER
Web Dashboard + WhatsApp Bot + Telegram Bot — Voice (OceanVoice) + Text Chat + Query Suggestions

AI BRAIN LAYER
Intent Detection → NL-to-SQL (LLM) → RAG on Metadata (FAISS) → Validation + AnomalyRadar Engine

RESPONSE COMPOSER
3D OceanLens Depth Profile + Interactive Map + Insight Text + Voice Response + CSV/ASCII Export

PoC Demo Scope: Indian Ocean Argo data (INCOIS/GDAC) • 3D depth profiles, trajectories, time-series • Voice demo (Hindi+English) • WhatsApp bot live • AnomalyRadar alert demo

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FEASIBILITY AND VIABILITY

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Feasibility Analysis

- Data 100% available & free — Argo GDAC + INCOIS open data; argopy library streamlines access
- All technologies open-source / free-tier — zero licensing cost (LLaMA/Mistral via OmniRoute, Web Speech API browser-native)
- Proven pattern — text-to-SQL + RAG is established; we add a domain-specific ocean intelligence layer
- Team capability — React + Python + FastAPI stack already configured; PoC achievable in hackathon window
- INCOIS ecosystem fit — PFZ advisory already uses WhatsApp/SMS delivery; FloatChat extends the same channel with interactive AI

Challenges & Mitigation Strategy

- LLM generates wrong SQL (hallucination) → Schema-constrained prompting + SQL validation layer + strictly read-only execution
- NetCDF data volume too large → Indian Ocean subset for PoC; Parquet compression + indexed queries
- Ambiguous voice queries → Clarifying-question loop + query suggestion chips + intent confidence scoring
- LLM API cost/latency → Open-source models via OmniRoute auto-fallback + cached frequent queries
- WhatsApp API limits → Rate-limiting + queue system; Telegram as parallel delivery channel

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IMPACT AND BENEFITS

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Impact on Target Audience

- Fishermen (1 Crore+ registered with INCOIS): voice query in native language on WhatsApp — PFZ-style advisory, no literacy barrier, no app download
- Coastal Officers / INCOIS Scientists: proactive AnomalyRadar alerts → faster cyclone preparedness & marine heatwave response
- Researchers / Students: 3D OceanLens exploration — hours of NetCDF data-wrangling reduced to seconds of conversation
- Educators / Schools (10 Lakh+ coastal students): FloatChat Classroom “Adopt a Float” — live ocean science aligned with NEP 2020

Benefits

- Social: Ocean data democratization across literacy levels & languages; regional voice inclusion; experiential science education
- Economic: Fuel savings for fishermen (INCOIS PFZ saves est. ₹700 Cr/year); zero-cost open-source stack; 10x research productivity
- Environmental: Early marine heatwave & cyclone precursor detection → faster climate adaptation; ecosystem health monitoring
- National: Direct INCOIS / MoES / Blue Economy Mission alignment; potential integration into the SAMUDRA app ecosystem

“If INCOIS adopts FloatChat as its public-facing AI layer, 1 Crore+ registered fishermen get interactive ocean intelligence on their existing WhatsApp — no new infrastructure, no training cost, no literacy barrier.”

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RESEARCH AND REFERENCES

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- International Argo Program — GDAC: argodatamgt.org / argo.ucsd.edu
- INCOIS (MoES, Govt. of India) — Indian Argo Project: incois.gov.in
- Argo float data — NetCDF specification & free access: doi.org/10.17882/42182
- Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks” (NeurIPS 2020)
- IndOOS (Indian Ocean Observing System) Decadal Review — CLIVAR: clivar.org
- INCOIS Potential Fishing Zone (PFZ) Advisory Service — WhatsApp/SMS dissemination framework
- argopy: Python library for Argo data access — argopy.readthedocs.io
- NEP 2020 — National Education Policy, experiential learning mandate (Ministry of Education, Govt. of India)